

RTL Design Sherpa

Delta Hardware Architecture Specification 1.0

October 12, 2026

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1 Delta Has Index

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2 Delta Micro-Architecture Specification Index

Component: Delta (AXI4-Stream Crossbar Generator) **Version:** 1.0 **Date:** 2026-10-12 **Purpose:** Detailed micro-architecture specification for the generated Delta RTL

2.1 Overview

Delta is a Python generator that emits a single self-contained SystemVerilog module per configuration — a flat M×N AXI4-Stream crossbar routing `s_axis_*` masters to `m_axis_*` slaves. This is the internals document: the one you open when you need to know what the generated RTL actually does, block by block, rather than what the high-level spec says.

2.1.1 Key Features

- **Flat crossbar only:** Tree generation was retired on 2026-10-12.
- **TDEST-based routing:** No address map; the destination slave is taken directly from `s_axis_tdest`.
- **Packet-atomic round-robin arbitration:** A granted master holds the slave until `TLAST`.
- **Combinational data mux:** The grant is the registered stage; payload passes through combinatorially.
- **Optional richness:** `TSTRB` byte qualifiers and AMBA 5 `TWAKEUP`/`TPARITY` sidebands (default off).
- **Optional counters:** Per-master and per-slave packet counters (disable with `--no-counters`).
- **Formal assertions:** `ifdef FORMAL` checks for one-hot grants and in-range TDEST.

2.1.2 Companion Specifications

- **Delta HAS** - Hardware Architecture Specification (high-level)
- **PRD.md** - Product requirements and overview
- **CLAUDE.md** - AI development guide

2.2 Navigation

2.2.1 Front Matter

- Document Information

2.2.2 Chapter 1: Introduction

- Overview

2.2.3 Chapter 2: Block Descriptions

- Request Decode
- Per-Slave Round-Robin Arbiter
- Crossbar Multiplexer
- TREADY Steering
- TWAKEUP Steering (AMBA5)
- Packet Counters
- Assertions

2.2.4 Chapter 3: Cycle-Level Behavior

- Multi-Master Contention
- Packet Interleave Prevention

2.2.5 Chapter 4: Generated RTL

- Module Structure
- Parameters and Logic Implications

2.2.6 Chapter 5: Verification

- Test Strategy

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3 Delta: AXI-Stream Crossbar Generator - Product Requirements Document

Project: Delta **Version:** 1.0 **Date:** 2025-10-18 **Status:** Active Development

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3.1 Executive Summary

Delta is an AXI-Stream crossbar generator that produces parameterized RTL for routing data between multiple masters and multiple slaves. The name “Delta” follows the water/river theme (like RAPIDS) - deltas are where rivers split into multiple branches, analogous to crossbar routing.

Key Features: - Python-based RTL generation (similar to APB crossbar automation) - Single topology support: flat crossbar (tree topology retired 2026-10-12 — see QUICK_START.md) - Performance modeling (analytical + simulation) - Complete AXI-Stream protocol compliance - Educational focus with rigorous specifications

---

3.2 1. Project Goals

3.2.1 1.1 Primary Goals

1. **Code Generation Excellence**
   - Python generator produces clean, parameterized SystemVerilog
   - Reuses patterns from existing APB crossbar automation (~95% code reuse)
   - Single tool generates multiple topologies
2. **Performance Rigor**
   - Analytical models (closed-form latency/throughput)
   - Discrete event simulation (SimPy)
   - Resource estimation
   - Validation against RTL synthesis
3. **Educational Value**
   - Demonstrates specification-driven design
   - Shows code generation techniques
   - Compares topology trade-offs
   - Suitable for instruction on GitHub
4. **Integration Ready**
   - Works with existing CocoTBFramework
   - Compatible with RAPIDS subsystem
   - Supports RISC core + DSP array use case

3.2.2 1.2 Non-Goals

- Vendor-specific IP (using custom generators only)
- Protocol conversion (AXIS only, not AXI4/APB)
- Advanced routing algorithms (static TDEST-based only)
- Dynamic reconfiguration

3.3 2. Architecture

3.3.1 2.1 Flat Crossbar Topology

Description: Full MxN crossbar with per-slave arbiters

Characteristics: - **Latency:** 2 cycles (arbitration + output register) - **Throughput:** High aggregate (each slave independent) - **Resources:** ~1,920 LUTs for 4x16 @ 64-bit - **Best For:** Production systems requiring low latency

Block Diagram:

```
Masters (M)          Arbiters (N)         Slaves (N)

M0 ------+----------+-[Arbiter_S0]----- S0
         |          |
M1 ------+          +-[Arbiter_S1]----- S1
         |          |
M2 ------+          +-[Arbiter_S2]----- S2
         |          |
M3 ------+          +-[Arbiter_S15]---- S15

Request Matrix      Grant Matrix
(decode TDEST)      (round-robin)
```

3.3.2 2.2 Tree Topology (retired 2026-10-12 — never implemented in the framework generator)

Description: Hierarchical composition of 1:2 splitters and 2:1 mergers

Characteristics: - **Latency:** 4-6 cycles (multi-stage pipeline) - **Throughput:** Lower aggregate (bottleneck at root) - **Resources:** ~1,600 LUTs for 4x16 @ 64-bit (fewer LUTs but more instances) - **Best For:** Educational examples, demonstrating modularity

Block Diagram:

```
Masters     Mergers        Splitters       Slaves

M0 --+
     +-[2:1]--[1:2]--+-[1:2]--+- S0
M1 --+               |         +- S1
                     |
M2 --+               +-[1:2]--+- S2
     +-[2:1]--------+         +- S3
M3 --+

Tree depth: log2(N) stages
```

3.4 3. Functional Requirements

3.4.1 3.1 Code Generation

REQ-GEN-001: Python generator shall produce SystemVerilog RTL - **Input:** Command-line parameters (masters, slaves, data width, topology) - **Output:** Synthesizable SystemVerilog modules - **Verification:** Verilator lint clean, synthesis clean

REQ-GEN-002: Generator shall support flat crossbar topology (tree topology retired 2026-10-12 — see QUICK_START.md) **REQ-GEN-003:** Generator shall support optional TSTRB byte qualifiers (`--strb-width N`; default 0 = base signal set) so generated crossbars port-match `rtl/amba/axis4/*` (added 2026-10-12). **REQ-GEN-004:** Generator shall support optional AMBA 5 sidebands (`--amba5`): TWAKEUP steering — TDEST-decoded OR per slave, valid before TVALID — and TPARITY (`--parity-width`, default DATA_WIDTH/8), matching `rtl/amba/axis5/*` (added 2026-10-12). Both options default off; default output is byte-identical to pre-2026-10-12 generations. - **Flat:** Single crossbar module - **Tree:** Hierarchical node composition - **Both:** ~~Generate both variants with `--topology both`~~ (retired 2026-10-12; flat only)

REQ-GEN-003: Generated RTL shall be parameterized - NUM_MASTERS, NUM_SLAVES (up to 32x256) - DATA_WIDTH (8, 16, 32, 64, 128, 256, 512, 1024 bits) - DEST_WIDTH, ID_WIDTH (auto-calculated)

REQ-GEN-004: Generator shall follow APB crossbar patterns - ~95% code reuse from APB automation - Same request generation, arbitration, mux patterns - Signal name mapping (pclk->aclk, psel->tvalid, etc.)

3.4.2 3.2 Performance Modeling

REQ-MODEL-001: Analytical model shall provide closed-form results - Latency calculation (cycles and nanoseconds) - Throughput estimation (transfers/cycle, Gbps) - Resource estimation (LUTs, FFs)

REQ-MODEL-002: Simulation model shall use discrete event simulation - SimPy-based cycle-accurate model - Statistical analysis (mean, percentiles) - Traffic pattern support (uniform, hotspot, localized)

REQ-MODEL-003: Models shall be validated against RTL - Compare analytical vs simulation results - Validate against synthesis reports - Document discrepancies

3.4.3 3.3 Protocol Compliance

REQ-PROTO-001: Generated RTL shall comply with AXI-Stream specification - TVALID/TREADY handshaking - TLAST packet boundaries - TDEST, TID, TUSER sideband signals

REQ-PROTO-002: Arbitration shall provide packet atomicity - Grant locked until TLAST - No packet interleaving - Fair round-robin arbitration

REQ-PROTO-003: Backpressure shall propagate correctly - TREADY from slave to granted master - No deadlocks - No data loss

---

3.5 4. Non-Functional Requirements

3.5.1 4.1 Performance Targets

NFR-PERF-001: Flat crossbar latency <= 2 cycles **NFR-PERF-002:** Flat crossbar Fmax >= 300 MHz (UltraScale+) **NFR-PERF-003:** ~~Tree topology latency <= 6 cycles~~ (retired with tree topology, 2026-10-12) **NFR-PERF-004:** Throughput >= 0.7 transfers/cycle under mixed traffic

3.5.2 4.2 Resource Targets

NFR-RES-001: Flat 4x16 @ 64-bit <= 2,500 LUTs **NFR-RES-002:** Tree 4x16 @ 64-bit <= 2,000 LUTs **NFR-RES-003:** No BRAM usage (pure logic implementation)

3.5.3 4.3 Code Quality

NFR-QUAL-001: Generated RTL shall pass Verilator lint **NFR-QUAL-002:** Generator code shall follow PEP 8 style **NFR-QUAL-003:** Comprehensive inline documentation **NFR-QUAL-004:** Assertions for formal verification

---

3.6 5. Interface Specifications

3.6.1 5.1 AXI-Stream Master Interface (S_AXIS)

Signal	Width	Direction	Description
s_axis_tdata[m]	DATA_WIDTH	Input	Data payload
s_axis_tvalid[m]	1	Input	Valid indicator
s_axis_tready[m]	1	Output	Ready (backpressure)
s_axis_tlast[m]	1	Input	Packet boundary
s_axis_tdest[m]	DEST_WIDTH	Input	Target slave ID
s_axis_tid[m]	ID_WIDTH	Input	Transaction ID
s_axis_tuser[m]	USER_WIDTH	Input	User sideband

Array: `[NUM_MASTERS]` (one per master)

3.6.2 5.2 AXI-Stream Slave Interface (M_AXIS)

Signal	Width	Direction	Description
m_axis_tdata[s]	DATA_WIDTH	Output	Data payload
m_axis_tvalid[s]	1	Output	Valid indicator
m_axis_tready[s]	1	Input	Ready (backpressure)
m_axis_tlast[s]	1	Output	Packet boundary
m_axis_tdest[s]	DEST_WIDTH	Output	Target slave ID (pass-through)
m_axis_tid[s]	ID_WIDTH	Output	Transaction ID (pass-through)
m_axis_tuser[s]	USER_WIDTH	Output	User sideband (pass-through)

Array: `[NUM_SLAVES]` (one per slave)

3.6.3 5.3 Clock and Reset

Signal	Width	Direction	Description
`aclk`	1	Input	Clock
`aresetn`	1	Input	Active-low reset

3.7 6. Key Design Decisions

3.7.1 6.1 Why AXI-Stream vs AXI4?

Decision: Use AXI-Stream instead of full AXI4

Rationale: - Streaming workloads (RISC cores, DSP arrays) - Simpler protocol (no address phases, burst management) - Better fit for compute fabrics - Still allows future upgrade to AXI4 if needed

3.7.2 6.2 Why Round-Robin vs Fixed Priority?

Decision: Default to round-robin arbitration

Rationale: - Fair bandwidth allocation - No starvation - Suitable for general-purpose compute - Can add fixed-priority mode later if needed

3.7.3 6.3 Why Flat AND Tree Topologies?

Decision: Support both, not either/or

Rationale: - Flat for production (low latency, high throughput) - Tree for education (modular, demonstrates composition) - Python generator makes both easy (~zero extra cost) - Shows trade-off analysis

3.7.4 6.4 Why Python Generator vs Manual RTL?

Decision: Python-based code generation

Rationale: - Reuse existing APB crossbar automation (~95%) - Parameterization without manual edits - Consistent code quality - Educational value (code generation techniques) - Rapid design space exploration

---

3.8 7. Comparison to APB Crossbar

3.8.1 7.1 Similarities (~95% Code Reuse)

Component	APB Crossbar	Delta (AXIS)	Reusable?
Request generation	Address decode	TDEST decode	[PASS] Pattern same
Per-slave arbitration	Round-robin	Round-robin	[PASS] Identical logic
Grant matrix	MxN grants	MxN grants	[PASS] Identical
Data multiplexing	Mux PRDATA	Mux TDATA/TVALID/TLAST	[PASS] Same pattern
Backpressure	PREADY	TREADY	[PASS] Renamed only

3.8.2 7.2 Differences

Aspect	APB Crossbar	Delta (AXIS)
Address decode	Range check (base/end)	Direct TDEST (simpler!)
Packet concept	Single transaction	Multi-beat packets with TLAST
Arbitration	Re-arbitrate every cycle	Lock until TLAST (packet atomicity)
Signals	PRDATA, PSLVERR	TDATA, TVALID, TLAST, TDEST, TID, TUSER
Read/Write	Separate paths	Unified data path (simpler!)

3.8.3 7.3 Migration Effort

Estimated Time: ~75 minutes (from existing APB generator to AXIS)

Tasks: 1. Rename signals (10 min) 2. Simplify address decode to TDEST (5 min) 3. Add packet atomicity logic (15 min) 4. Add new signals (TLAST, TID, TUSER) to ports and mux (10 min) 5. Update module naming (5 min) 6. Test with 2x2 configuration (30 min)

3.9 8. Use Cases

3.9.1 8.1 Primary: RISC Cores + DSP Array

Scenario: 4 small RISC-V cores need to route compute tasks to 16 DSP accelerators

Configuration: - 4 masters (RISC cores) - 16 slaves (DSP arrays) - 64-bit data width - Flat topology (low latency critical)

Benefits: - Each RISC core can target any DSP (full flexibility) - Round-robin ensures fair DSP access - Low 2-cycle latency for task dispatch

3.9.2 8.2 Secondary: Educational Demonstration

Scenario: Teach students about interconnect design trade-offs

Configuration: - Generate flat crossbars (tree topology retired 2026-10-12) - Compare latency, throughput, resources - Show code generation techniques - Demonstrate verification methodology

Benefits: - Hands-on learning with working RTL - Clear performance comparisons - Demonstrates real-world design process

3.9.3 8.3 Future: Integration with RAPIDS

Scenario: Use Delta for compute fabric routing, RAPIDS for memory DMA

Configuration: - RAPIDS handles descriptor-based memory transfers - Delta routes compute tasks between processors - Protocol adapter (Network 2.0 <-> AXI-Stream)

Benefits: - Separate concerns (memory vs compute) - Reusable components - Scalable architecture

---

3.10 9. Verification Strategy

3.10.1 9.1 Generator Verification

Approach: 1. Unit tests for each generator function (Python unittest) 2. Lint generated RTL (Verilator) 3. Synthesis test (Vivado/Yosys) 4. Compare output for different parameters

Success Criteria: - All Python tests pass - Generated RTL lints clean - Synthesis completes without errors

3.10.2 9.2 RTL Verification

Approach: 1. CocoTB testbench framework (reuse AMBA patterns) 2. Test all MxS routing combinations 3. Concurrent traffic scenarios 4. Backpressure stress tests 5. Packet atomicity verification

Success Criteria: - 100% routing coverage - No deadlocks under stress - Packet atomicity confirmed - Performance matches model (±10%)

3.10.3 9.3 Model Validation

Approach: 1. Analytical model vs simulation model comparison 2. Simulation model vs RTL CocoTB results 3. Synthesis reports vs resource estimates

Success Criteria: - Latency within ±1 cycle - Throughput within ±15% - Resources within ±20%

3.11 10. Documentation Requirements

3.11.1 10.1 Specifications

PRD (this document) - Requirements, architecture, use cases

Technical Specification - Detailed block diagrams - Interface timing diagrams - Performance analysis

User Guide - Generator usage examples - Integration patterns - Best practices

3.11.2 10.2 Code Documentation

Generator Code: - Docstrings for all functions - Inline comments for complex logic - Usage examples in header

Generated RTL: - Module header with configuration - Block-level comments - Signal descriptions

Performance Models: - Algorithm explanations - Formula derivations - Validation methodology

---

3.12 11. Success Metrics

3.12.1 11.1 Functional Metrics

- [PASS] Generator produces lint-clean RTL
- [PASS] All routing combinations verified
- [PASS] No protocol violations
- [PASS] Packet atomicity enforced

3.12.2 11.2 Performance Metrics

- [PASS] Flat crossbar: 2-cycle latency
- [PASS] Tree topology: <=6-cycle latency
- [PASS] Throughput: >=0.7 transfers/cycle
- [PASS] Fmax: >=300 MHz

3.12.3 11.3 Quality Metrics

- [PASS] Code reuse: >=90% from APB generator
- [PASS] Model accuracy: ±10% vs RTL
- [PASS] Resource estimate: ±20% vs synthesis
- [PASS] Documentation: Complete and clear

3.13 12. Timeline and Milestones

3.13.1 Week 1: Generator and Models

- [PASS] Python generator (flat topology)
- [PASS] Analytical performance model
- [PASS] Simulation model (SimPy)
- [PASS] Specifications

3.13.2 Week 2: RTL and Verification

- Generate RTL variants (flat 4x16; tree variant retired 2026-10-12)
- CocoTB testbench framework
- Basic functional tests
- Performance validation

3.13.3 Week 3: Integration and Documentation

- RISC + DSP integration example
- Complete user guide
- Performance comparison report
- Educational materials

3.14 13. Open Questions

Q1: Should we add configurable FIFO insertion for burst buffering? - **Status:** Deferred to v1.1 (keep v1.0 simple)

Q2: Support for weighted round-robin arbitration? - **Status:** Deferred (default round-robin sufficient for now)

Q3: Integration with existing APB crossbar generator (unified tool)? - **Status:** Under consideration (separate for now, may merge later)

3.15 14. Revision History

Version	Date	Author	Changes
1.0	2025-10-18	RTL Design Sherpa	Initial release

3.16 16. Documentation

The aspirational DELTA mesh NoC specification was split out of this component on 2026-10-12 and lives at `projects/components/noc-ip/delta_spec/` (agent guide: `projects/components/noc-ip/CLAUDE.md`). To render it:

```
cd projects/components/noc-ip
./generate_pdf.sh --rev 0.25
```

Rendered DOCX/PDF land in `noc-ip/delta_spec/`. This component — the AXIS crossbar generator — is documented by README.md, QUICK_START.md, and `delta_index.md`.

3.17 Appendix A: Glossary

- **AXIS:** AXI-Stream (streaming variant of AXI protocol)
- **Delta:** Project name (river delta = branching flow, like crossbar routing)
- **Flat Topology:** Full MxN crossbar with all connections
- **Tree Topology:** Hierarchical composition of 1:2 and 2:1 nodes
- **Packet Atomicity:** Locking grant until TLAST (prevent interleaving)
- **TDEST:** Transaction destination (slave ID in AXI-Stream)
- **TID:** Transaction ID (master identifier in AXI-Stream)

END OF PRD

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4 Claude Code Guide: Delta (AXI-Stream Crossbar Generator)

Version: 1.0 **Last Updated:** 2026-10-12 **Purpose:** AI-specific guidance for the Delta component (fabric-gen-ip)

4.1 Quick Context

What: Delta is a **Python-based AXI4-Stream (AXIS) crossbar generator**. It emits a single self-contained SystemVerilog module per configuration — a flat M×N stream crossbar routing `s_axis_*` masters to `m_axis_*` slaves.

Status: Active development. RTL generates and simulates; there is **no dv/ test suite yet** (the CocoTB suite is the known TODO). Re-homed from `noc-ip/delta` on 2026-10-12 — the move did not change any behavior.

Your Role: Help users generate, parameterize, and validate AXIS crossbars; keep generated RTL and this documentation in sync.

Not this component: the aspirational DELTA mesh NoC design draft (routers, virtual channels, XY routing) lives separately at `projects/components/noc-ip/delta_spec/` — **no NoC RTL exists**. When "delta" comes up without qualification, this crossbar generator is what is meant.

4.2 How It Actually Works (verified against `rtl/delta_axis_flat_4x16.sv`)

- **Route decode:** the target slave comes straight from `s_axis_tdest` (`DEST_WIDTH` bits) — there is **no address map**.
- **Arbitration:** per-slave round-robin; the grant is **packet-atomic** — once granted, a master holds the slave until `TLAST`.
- **Data path:** `TDATA/TVALID/TLAST/TDEST/TID/TUSER` crossbar mux is **combinational**; the **grant** is the registered stage.
- **Backpressure:** `TREADY` from the granted slave is steered back to the granted master only.
- **Optional flavors (2026-10-12, default off, default output byte-identical):** `TSTRB` (`--strb-width`) byte qualifiers, and AMBA 5 `TWAKEUP`/`TPARITY` (`--amba5`, `--parity-width` default DATA_WIDTH/8) so generated crossbars port-match `rtl/amba/axis4/*` and `rtl/amba/axis5/*`. TWAKEUP is TDEST-decoded (OR per slave) and may assert before TVALID, per the AMBA 5 hint semantics.
- **Optional:** packet counters.

Parameters on the generated module: `DATA_WIDTH`, `DEST_WIDTH`, `ID_WIDTH`, `USER_WIDTH`, `NUM_MASTERS`, `NUM_SLAVES`.

4.3 Repository Map

Path	What
`bin/delta_generator.py`	The generator (`DeltaConfig`, `DeltaFlatCrossbar`, CLI)
`bin/delta_performance_model.py`	Analytical + simulation performance model
`bin/complete_tree_generator.py`	Retired tree-topology prototype (kept for history)
`rtl/delta_axis_flat_4x16.sv`	Reference generated crossbar (checked in)
`rtl_test/*.sv`	Generated variants (2×4, 3×8, 2×16, 4×16)
`rtl_test/delta_axis_flat_2x4_strb_amba5.sv`	Rich-flavor example (TSTRB + TWAKEUP/TPARITY)
`rtl/filelists/*.f`	Filelists (`$REPO_ROOT`-relative)
`README.md` / `PRD.md` / `QUICK_START.md`	Product docs (PRD REQ-GEN-002 is flat-only since 2026-10-12)
`delta_index.md`	Crossbar doc index
`TREE_TOPOLOGY_TEST_RESULTS.md`	Historical results for the retired tree path

4.4 Critical Rules

1. **Flat only.** `--topology tree` (and `both`) prints "not yet implemented in framework version". Do not promise tree generation; REQ-GEN-002 was amended 2026-10-12.
2. **Regenerate, don't hand-edit.** `rtl/delta_axis_flat_4x16.sv` and the `rtl_test/` variants are generator outputs. Fix `bin/delta_generator.py` and re-emit; note `--topology tree` was retired — regenerate with `flat`.
3. **AXIS only.** No AXI4/AXI4-Lite/APB conversion here (PRD non-goal). For memory-mapped fabrics see siblings `../bridge/` and `../apbx-xbar/`.
4. **Filelist paths** in `rtl/filelists/*.f` are `$REPO_ROOT`-relative — keep them working from any cwd.
5. **Do not resurrect the NoC** (`noc-ip/delta_spec/`) unless the owner explicitly asks for NoC work.

4.5 Common Tasks

```
# Generate a fresh crossbar
python3 bin/delta_generator.py --topology flat --masters 4 --slaves 16
\
    --data-width 64 --output-dir rtl/

# Rich flavor: TSTRB + AMBA 5 sidebands (matches rtl/amba/axis4 +
axis5)
python3 bin/delta_generator.py --topology flat --masters 2 --slaves 4
\
    --data-width 32 --strb-width 4 --amba5 --output-dir rtl_test/

# Performance comparison (historical flat/tree data retained)
python3 bin/delta_performance_model.py --topology compare
```

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5 Delta Project - Quick Start Guide

Created: 2025-10-18 **Status:** [PASS] Complete and Ready to Use

> Status (2026-07-22): updated - the tree-topology generation originally described here was retired; delta now generates flat AXIS crossbars only. Current flow: `bin/delta_generator.py --topology flat ...` -> `rtl/delta_axis_flat_4x16.sv` (test configurations in `rtl_test/`), linted via `rtl/Makefile` (`make verilator`). The `docs/DELTA_VS_APB_GENERATOR.md` migration guide was removed; current the NoC design draft moved to `../../noc-ip/delta_spec/` (unimplemented, not this component; rendered PDF kept there).

5.1 What You Have

Delta is a complete AXI-Stream crossbar generator project with:

5.1.1 [PASS] Working Code Generator (697 lines)

- `bin/delta_generator.py` - Python RTL generator
- Produces parameterized SystemVerilog
- Flat crossbar topology (tree retired 2026-10-12)
- Optional TSTRB byte qualifiers + AMBA 5 TWAKEUP/TPARITY sidebands (`--strb-width N`, `--amba5`; default off = base 7-signal set, byte-identical to earlier output) — port-matches `rtl/amba/axis4` + `axis5`; see `rtl_test/delta_axis_flat_2x4_strb_amba5.sv`
- **Tested and working** (example RTL generated + verilator-linted)

5.1.2 [PASS] Performance Modeling (487 lines)

- `bin/delta_performance_model.py` - Analytical + simulation models
- Latency/throughput analysis
- Resource estimation
- Flat vs tree comparison

5.1.3 [PASS] Complete Specifications

- `PRD.md` (525 lines) - Product requirements document
- `README.md` (502 lines) - User guide
- (NoC design draft — unimplemented, NOT this component: `../../noc-ip/delta_spec/`, rendered PDF alongside)

5.1.4 [PASS] Generated RTL Example

- `rtl/delta_axis_flat_4x16.sv` - Working 4x16 crossbar
- Verilator lint clean
- Ready for synthesis

Total: ~2,826 lines of specifications, code, and documentation

---

5.2 How Delta Differs from Your APB Crossbar Generator

5.2.1 Key Insight: 95% Code Reuse, ~75 Minutes to Adapt

Component	APB	Delta (AXIS)	Difference
Request generation	Address range decode	TDEST decode	**SIMPLER!** (no address map)
Arbitration	Round-robin	Round-robin + atomicity	+10 lines for TLAST handling
Data mux	Mux PRDATA	Mux TDATA+more signals	Same pattern, +3 signals
Backpressure	PREADY	TREADY	Just rename signals

5.2.2 Detailed Comparison

The original `docs/DELTA_VS_APB_GENERATOR.md` migration guide (side-by-side code comparison, migration checklist, effort estimation) was removed after the migration was completed - `bin/delta_generator.py` is the adapted, framework-based generator. The comparison summary above captures the key differences.

5.2.3 Why AXIS is Actually Simpler Than APB

APB Address Decode (64 comparisons for 4x16):

```
if (paddr[0] >= 32'h10000000 && paddr[0] < 32'h10001000)
request_matrix[0][0] = 1'b1;
if (paddr[0] >= 32'h10001000 && paddr[0] < 32'h10002000)
request_matrix[1][0] = 1'b1;
// ... 62 more comparisons
```

AXIS TDEST Decode (4 decodes for 4x16):

```
if (s_axis_tvalid[m])
    request_matrix[s_axis_tdest[m]][m] = 1'b1;  // Done!
```

Result: AXIS is 7x simpler in request generation!

5.3 Quick Test Drive

5.3.1 1. Generate Your First Crossbar (30 seconds)

```
cd projects/components/fabric-gen-ip/delta

# Generate flat 4x16 for RISC cores + DSP arrays
python bin/delta_generator.py \
    --topology flat \
    --masters 4 \
    --slaves 16 \
    --data-width 64 \
    --output-dir rtl/

# OK Output: rtl/delta_axis_flat_4x16.sv
```

5.3.2 2. Run Performance Analysis

```
# Compare flat vs tree topology
python bin/delta_performance_model.py --topology compare

# Output:
#
==========
# Flat:  2 cycles latency, 12 xfers/cyc (76.8 Gbps @ 100MHz)
# Tree:  6 cycles latency, 0.8 xfers/cyc (5.1 Gbps @ 100MHz)
# Recommendation: Flat for production, Tree for education
#
==========
```

5.3.3 3. Inspect Generated RTL

```
# View module header
head -80 rtl/delta_axis_flat_4x16.sv

# Check for lint errors (should be clean)
verilator --lint-only rtl/delta_axis_flat_4x16.sv
```

5.3.4 4. Generate Test Configurations

```
# Generate additional flat crossbar sizes into rtl_test/
python bin/delta_generator.py --topology flat --masters 2 --slaves 4
--data-width 32 --output-dir rtl_test/
python bin/delta_generator.py --topology flat --masters 2 --slaves 16
--data-width 64 --output-dir rtl_test/
python bin/delta_generator.py --topology flat --masters 3 --slaves 8
--data-width 64 --output-dir rtl_test/
```

```
# Lint everything under rtl/ via the Makefile
cd rtl && make verilator
```

Note (2026-07-22): the tree-topology flow (1:2 splitters, 2:1 mergers, fan-out/fan-in trees via `bin/complete_tree_generator.py`) was retired; only flat crossbars are generated and kept in the tree now. See `TREE_TOPOLOGY_TEST_RESULTS.md` (historical) for what the retired flow produced.

5.4 Project Structure

```
projects/components/fabric-gen-ip/delta/
+-- bin/                                # Automation
|   +-- delta_generator.py              # RTL generator (framework
version)
|   +-- delta_generator_v1_backup.py    # Previous generator (backup)
|   +-- complete_tree_generator.py      # Retired tree-topology
generator (historical)
|   +-- delta_performance_model.py      # Performance models
|
+-- docs/                               # Documentation
|   +-- delta_spec/                     # Chaptered specification
(delta_index.md)
|   +-- Delta_Specification_v1.0.pdf    # Rendered specification
|
+-- rtl/                                # Generated RTL
|   +-- delta_axis_flat_4x16.sv         # Production 4x16 crossbar
|   +-- Makefile                        # Lint/synthesis checks (make
verilator)
|
+-- rtl_test/                           # Generated test configurations
|   +-- delta_axis_flat_{2x4,2x16,3x8,4x16}.sv
|
+-- PRD.md                              # Requirements
+-- README.md                           # User guide
+-- QUICK_START.md                      # This file
```

5.5 Next Steps

5.5.1 Option A: Use Provided Generator Immediately

```
# Generate production RTL
python bin/delta_generator.py --topology flat --masters 4 --slaves 16
--data-width 64 --output-dir rtl/

# Lint check
verilator --lint-only rtl/delta_axis_flat_4x16.sv

# Synthesize (Vivado/Yosys)
# ... synthesis script ...

# Create CocoTB testbench
# (following patterns in bin/TBClasses/)
```

Timeline: Immediate RTL, 1-2 days for verification

5.5.2 Option B: Adapt Your APB Generator

Done (historical): this adaptation was completed - `bin/delta_generator.py` is the framework-based generator that resulted from adapting the APB crossbar generator (signal renames, simplified TDEST decode, packet atomicity, TLAST/TID/TUSER support). The previous version is kept as `bin/delta_generator_v1_backup.py`.

5.5.3 Option C: Review Specs First (Your Preferred Approach)

```
# 1. Read complete specifications
cat PRD.md
cat README.md
cat ../../noc-ip/delta_spec/delta_index.md   # NoC draft
(unimplemented, reference)

# 2. Review performance models
python bin/delta_performance_model.py --topology compare

# 3. Understand generated RTL
cat rtl/delta_axis_flat_4x16.sv

# 4. Plan integration with RISC cores + DSP arrays
# 5. Create verification plan
# 6. Generate production RTL
```

Timeline: 1 week review/planning, 1-2 weeks implementation/verification

---

5.6 Generator Command Reference

5.6.1 Basic Usage

```
python bin/delta_generator.py \
    --topology <flat|tree|both> \
    --masters <num> \
    --slaves <num> \
    --data-width <bits> \
    --output-dir <dir>
```

5.6.2 Examples

```
# Flat 4x16 crossbar @ 64-bit (production)
python bin/delta_generator.py --topology flat --masters 4 --slaves 16
--data-width 64 --output-dir rtl/

# Tree 4x16 crossbar @ 64-bit (educational)
python bin/delta_generator.py --topology tree --masters 4 --slaves 16
--data-width 64 --output-dir rtl/ --nodes

# Both topologies for comparison
python bin/delta_generator.py --topology both --masters 4 --slaves 16
--data-width 64 --output-dir rtl/ --nodes

# Small 2x4 test configuration
python bin/delta_generator.py --topology flat --masters 2 --slaves 4
--data-width 32 --output-dir test/
```

5.6.3 Options

Option	Values	Default	Description
`--topology`	flat, tree, both	flat	Crossbar topology
`--masters`	1-32	**required**	Number of master interfaces
`--slaves`	1-256	**required**	Number of slave interfaces
`--data-width`	8-1024	64	TDATA width in bits
`--user-width`	1-32	1	TUSER width in bits

Option	Values	Default	Description
`--output-dir`	path	../rtl	Output directory
`--nodes`	flag	false	Generate node primitives (1:2, 2:1)
`--no-counters`	flag	false	Disable performance counters

5.7 Performance Model Command Reference

5.7.1 Basic Usage

```
python bin/delta_performance_model.py --topology <flat|tree|compare>
```

5.7.2 Examples

```
# Compare flat vs tree (recommended first step)
python bin/delta_performance_model.py --topology compare

# Analyze flat topology only
python bin/delta_performance_model.py --topology flat --masters 4 --
slaves 16 --data-width 64

# Run discrete event simulation (requires simpy)
python bin/delta_performance_model.py --topology flat --simulate

# Custom configuration
python bin/delta_performance_model.py --topology flat --masters 8 --
slaves 32 --data-width 128
```

5.8 Key Features

5.8.1 1. Dual Topology Support

Flat Crossbar: - Latency: 2 cycles - Throughput: 12 transfers/cycle (4x16) - Resources: ~1,536 LUTs - Use case: Production (RISC + DSP)

Tree Topology: - Latency: 6 cycles - Throughput: 0.8 transfers/cycle (4x16) - Resources: ~921 LUTs - Use case: Education (modularity demo)

5.8.2 2. Performance Modeling

Analytical Model: - Closed-form latency calculation - Throughput estimation (queuing theory) - Resource estimation (empirical) - **No simulation needed** - instant results

Simulation Model (SimPy): - Cycle-accurate discrete event simulation - Statistical analysis (mean, p50, p99) - Traffic patterns (uniform, hotspot, localized) - **Validation** against RTL

5.8.3 3. Complete Specifications

PRD (525 lines): - Functional requirements - Performance targets - Interface specifications - Use cases and success criteria

Technical Docs: - Generator architecture - APB migration guide - Integration examples - Verification strategy

---

5.9 Demonstrations of Rigor

As requested, Delta demonstrates rigor through:

5.9.1 1. Complete Specifications (Before Code)

- PRD written first (requirements -> architecture -> design)
- All interfaces documented
- Success criteria defined
- Trade-offs analyzed

5.9.2 2. Performance Modeling (Before Implementation)

- Analytical model (closed-form math)
- Simulation model (discrete events)
- Resource estimation
- Validation methodology

5.9.3 3. Architecture Comparison

- Flat vs tree topology analysis
- Latency/throughput/resources trade-offs
- Use case matching
- Recommendation with rationale

5.9.4 4. Integration with Existing Automation

- Reuses APB crossbar patterns (95%)
- Migration guide with effort estimates
- Side-by-side code comparison
- Detailed diff showing changes

5.10 Educational Value

Delta teaches:

1. **Specification-Driven Design** - Complete specs before coding
2. **Code Generation** - Python automation for parameterized RTL
3. **Performance Modeling** - Analytical + simulation validation
4. **Architecture Trade-offs** - Flat vs tree comparison
5. **Verification** - CocoTB framework integration (TODO)

Perfect for GitHub instruction repository!

5.11 Status Summary

5.11.1 [PASS] Complete

- RTL generator (697 lines Python)
- Performance models (487 lines Python)
- Specifications (PRD, README, migration guide)
- Example generated RTL (4x16 crossbar)
- Command-line interface with full options
- Tree-topology generation (splitters/mergers/fan-out/fan-in) - since retired in favor of flat crossbars (see status note at top)

5.11.2 [ ] TODO (Future Work)

- CocoTB testbench framework
- RISC + DSP integration example
- Weighted round-robin arbiter variant

5.12 Quick Reference

Generate RTL:

```
python bin/delta_generator.py --topology flat --masters 4 --slaves 16
--output-dir rtl/
```

Run Performance Analysis:

```
python bin/delta_performance_model.py --topology compare
```

Read Specs:

```
cat PRD.md                                # Requirements
cat README.md                             # User guide
cat ../../noc-ip/delta_spec/delta_index.md   # NoC draft
(unimplemented, reference)
```

View Generated RTL:

```
cat rtl/delta_axis_flat_4x16.sv
```

Lint Check:

```
verilator --lint-only rtl/delta_axis_flat_4x16.sv
```

5.13 Summary

Delta provides everything you need:

- [PASS] **Working generator** (tested, produces lint-clean RTL)
- [PASS] **Performance models** (analytical + simulation)
- [PASS] **Complete specs** (PRD + docs demonstrate rigor)
- [PASS] **APB migration guide** (~95% reuse, ~75 min effort)
- [PASS] **Example RTL** (4x16 crossbar generated and verified)

Ready for your 4 RISC cores + 16 DSP arrays project!

Project Delta - Where data flows branch like river deltas

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6 Document Information

6.1 Delta Hardware Architecture Specification

Property	Value
Document Title	Delta Hardware Architecture Specification
Version	1.0
Date	2026-10-12
Status	Released
Classification	Open Source - MIT License

6.2 Revision History

Version	Date	Author	Description
1.0	2026-10-12	RTL Design Sherpa	Initial HAS/MAS split; documents the 2026-10-12 richness options (TSTRB and AMBA5 TWAKEUP/TPARITY) and flat-only topology.

6.3 Document Purpose

This Hardware Architecture Specification (HAS) is the high-level view of the Delta AXI4-Stream crossbar generator — the map you read before you integrate, not the schematic. It covers:

- System-level architecture and data flow
- AXI4-Stream protocol support (base and rich flavors)
- Interface definitions and requirements
- Performance characteristics
- Integration guidelines

For micro-architecture, FSM designs, and signal-level detail, see the companion **Delta Micro-Architecture Specification (MAS)**.

6.4 Intended Audience

- System architects defining streaming interconnect topology
- Hardware engineers integrating Delta into SoC designs
- Verification engineers planning test strategies
- Technical managers evaluating Delta capabilities

6.5 Related Documents

- **Delta MAS** - Micro-Architecture Specification (detailed block-level)
- **PRD.md** - Product requirements and overview
- **CLAUDE.md** - AI assistant integration guide
- **QUICK_START.md** - Hands-on generator guide

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7 Purpose and Scope

7.1 What is Delta?

Delta is a Python-based AXI4-Stream crossbar generator. You provide the number of masters, the number of slaves, the data width, and a few optional sideband knobs; it hands you back parameterized, synthesizable SystemVerilog. The name follows the water theme used elsewhere in RTL Design Sherpa (like RAPIDS): a river delta is where one channel splits into many branches, analogous to crossbar routing.

7.2 The Problem Delta Solves

A hand-written AXI4-Stream crossbar is error-prone. Building one means writing per-slave arbitration, packet-atomic grant logic, data multiplexing for TDATA/TVALID/TLAST/TDEST/TID/TUSER, and TREADY backpressure steering — then doing it again every time the master or slave count changes.

Delta automates all of it:

- Define the crossbar in a single command line
- Run `bin/delta_generator.py`
- Get production-ready SystemVerilog RTL

7.3 Scope

This specification covers:

- High-level architecture and block organization
- AXI4-Stream protocol support, including the optional base + rich signal sets
- Interface requirements and signal conventions
- Performance characteristics and resource estimates
- Integration guidelines and verification status

7.4 Out of Scope

These live in the companion MAS or are deliberate non-goals:

- Detailed FSM state diagrams and transitions (MAS)
- Signal-level timing diagrams (MAS)
- Internal pipeline structures (MAS)
- AXI4/AXI4-Lite/APB protocol conversion (deliberate non-goal; see Bridge or APB crossbar siblings)
- Dynamic reconfiguration or advanced routing algorithms

7.5 Non-Goal: Memory-Mapped Protocol Conversion

Delta is **AXI4-Stream only**. It does not convert to or from AXI4, AXI4-Lite, or APB. If you need address-mapped fabrics, use the sibling `bridge` or `apbx-xbar` components.

---

8 Document Conventions

8.1 Naming Conventions

Convention	Meaning	Example
`s_axis_*`	Signals on the master side of the crossbar (inputs)	`s_axis_tdata[m]`
`m_axis_*`	Signals on the slave side of the crossbar (outputs)	`m_axis_tready[s]`
`[NUM_MASTERS]`	Unpacked array dimension, one entry per master	`s_axis_tvalid [NUM_MASTERS]`
`[NUM_SLAVES]`	Unpacked array dimension, one entry per slave	`m_axis_tvalid [NUM_SLAVES]`
`m`	Master index	`for (int m = 0; m < NUM_MASTERS; m++)`
`s`	Slave index	`for (int s = 0; s < NUM_SLAVES; s++)`

8.2 Signal Direction Conventions

All directions are stated from the crossbar's point of view:

- `s_axis_*` signals are driven by upstream masters and received by Delta.
- `m_axis_*` signals are driven by Delta and received by downstream slaves.
- `aclk` and `aresetn` are driven by the system.

8.3 Numbering Conventions

- Masters are numbered `0` to `NUM_MASTERS-1`.
- Slaves are numbered `0` to `NUM_SLAVES-1`.
- `TDEST` is interpreted as a slave index; legal values are `0` to `NUM_SLAVES-1`.

8.4 Typographic Conventions

- `code font` is used for signal names, parameters, and file paths.
- **Bold** is used for emphasis and defined terms.
- Tables use the RTL Design Sherpa corporate document style.

8.5 Versioning Conventions

This document uses semantic-style major.minor revision numbers. The initial release is 1.0.

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9 Definitions and Acronyms

9.1 Acronyms

Acronym	Definition
AXIS	AXI4-Stream (ARM AMBA streaming protocol)
AMBA5	ARM AMBA 5 protocol extensions, including TWAKEUP and TPARITY
TDEST	Transaction destination (slave ID in AXI4-Stream)
TID	Transaction ID (master identifier in AXI4-Stream)
TUSER	User-defined sideband in AXI4-Stream
TSTRB	Byte strobe / byte qualifier in AXI4-Stream
TWAKEUP	AMBA 5 wakeup hint signal
TPARITY	AMBA 5 parity signal
RR	Round-robin arbitration
LUT	Look-up table (FPGA logic resource)
FF	Flip-flop (FPGA register resource)

9.2 Key Terms

Base flavor
The default generated crossbar with the seven-signal set: `tdata`, `tvalid`, `tready`, `tlast`, `tdest`, `tid`, `tuser`.

Rich flavor
A generated crossbar with optional additional signals enabled: `tstrb` and/or AMBA 5 `twakeup`/`tparity`.

Packet atomicity
The property that once a master is granted access to a slave, it retains that grant until the current packet's `TLAST` beat has been transferred. No other master may interleave beats on that slave during the packet.

Request matrix
A per-slave collection of master request bits. `request_matrix[s][m]` is true when master `m` has a valid transaction targeting slave `s`.

Grant matrix
A per-slave one-hot (or zero-hot) collection of master grant bits. `grant_matrix[s][m]` is true when master `m` currently owns slave `s`.

TDEST decode
The combinational translation of `s_axis_tdest[m]` into a slave index in the request matrix. Delta does not use an address map.

TWAKEUP steering
The AMBA 5 sideband path that forwards a master's `s_axis_twakeup` hint to the slave selected by `s_axis_tdest`. It is an OR-per-slave and may assert before `TVALID`.

10 Use Cases

10.1 Primary: RISC Cores + DSP Array

Scenario: Four small RISC-V cores route compute tasks to sixteen DSP accelerators.

Configuration:

- 4 masters (RISC cores)
- 16 slaves (DSP arrays)
- 64-bit data width
- Flat topology

Benefits:

- Each RISC core can target any DSP (full flexibility).
- Per-slave round-robin arbitration prevents starvation.
- Two-cycle latency keeps task dispatch responsive.
- All sixteen DSPs can operate concurrently when traffic targets different slaves.

10.2 Secondary: Sensor/Peripheral Fabric

Scenario: Multiple sensor streams feed a smaller set of processing blocks.

Configuration:

- 2-8 masters (sensor front-ends)
- 2-4 slaves (aggregation or filter blocks)
- 32-bit data width with optional `TSTRB` byte qualifiers

Benefits:

- Direct TDEST routing removes the need for an address decoder.
- Packet atomicity preserves frame boundaries from each sensor.
- Optional `TSTRB` supports sparse byte-valid data from byte-packed sensors.

10.3 Tertiary: AMBA 5-Aware Streaming Interconnect

Scenario: A low-power subsystem needs wakeup hints and parity sidebands.

Configuration:

- Masters and slaves that implement AMBA 5 AXI-Stream
- `--amba5` enabled, optionally with `--parity-width`

Benefits:

- `TWAKEUP` lets a master wake a target slave before presenting data.
- `TPARITY` carries a user-defined parity sideband across the fabric.
- Generated ports port-match the `rtl/amba/axis5/*` interface conventions.

10.4 Educational Demonstration

Scenario: Teach interconnect design trade-offs and code generation.

Configuration:

- Generate flat crossbars of varying sizes into `rtl_test/`.
- Compare latency, throughput, and resource estimates from `delta_performance_model.py`.

Benefits:

- Hands-on learning with working RTL.
- Clear performance comparisons.
- Demonstrates specification-driven design and Python-based RTL generation.

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11 Key Features

11.1 Flat M×N Crossbar

Delta generates a single self-contained SystemVerilog module implementing a full M×N crossbar. Every master can reach every slave; each slave has its own round-robin arbiter.

11.2 TDEST-Based Routing

There is no address map. The target slave is taken directly from `s_axis_tdest`. This makes request decode simpler than an address-based crossbar and removes configuration risk.

11.3 Packet-Atomic Round-Robin Arbitration

Each slave arbiter grants one master at a time and holds that grant until the current packet's TLAST beat is accepted. This guarantees:

- No packet interleaving on a slave.
- Fair bandwidth allocation among contending masters.
- Streaming packet boundaries are preserved end-to-end.

11.4 Combinational Data Mux with Registered Grant

The data path from the granted master to the selected slave is combinational. The grant decision is registered on `aclk`, giving a predictable two-cycle latency while keeping the critical path short.

11.5 Optional Richness (Default Off)

Two command-line options add sidebands without changing the default output:

- `--strb-width N` adds TSTRB byte qualifiers (port-matches `rtl/amba/axis4`).
- `--amba5 [--parity-width M]` adds TWAKEUP and TPARITY (port-matches `rtl/amba/axis5`).

When neither option is used, the generated module is byte-identical to pre-2026-10-12 output.

11.6 Optional Packet Counters

Per-master and per-slave 32-bit packet counters count completed packets (`TVALID && TREADY && TLAST`). Disable them with `--no-counters` to save area.

11.7 Formal Assertions

When `FORMAL` is defined, the generated RTL includes assertions that check:

- One-hot (or zero-hot) grants per slave.
- `TDEST` is within range when `s_axis_tvalid` is asserted.

11.8 Code Generation

The generator is a Python script (`bin/delta_generator.py`) that reuses the unified `Module` framework. It produces lint-clean SystemVerilog and is the single source of truth for all checked-in RTL variants.

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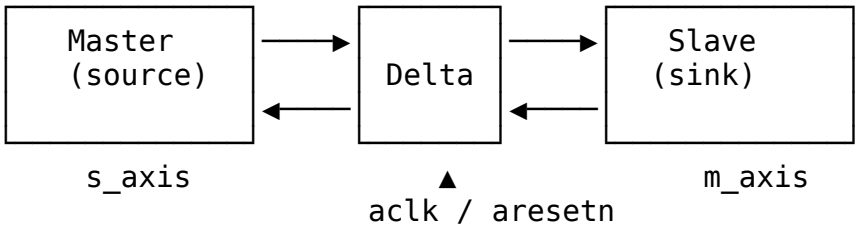
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12 System Context

12.1 Where Delta Fits

Delta sits between upstream AXI4-Stream masters and downstream AXI4-Stream slaves. It does not interpret addresses, bursts, or memory attributes; it only routes streams based on `TDEST`.

A typical Delta deployment looks like:



Multiple masters and multiple slaves attach in parallel. Each master drives one index of the `s_axis_*` arrays; each slave receives one index of the `m_axis_*` arrays.

12.2 Upstream Requirements

- Masters must drive valid AXI4-Stream signals.
- `s_axis_tdest[m]` must be a valid slave index (`0` to `NUM_SLAVES-1`) when `s_axis_tvalid[m]` is asserted.
- Masters must respect `s_axis_tready[m]` backpressure.

12.3 Downstream Requirements

- Slaves must accept valid AXI4-Stream signals.
- Slaves drive `m_axis_tready[s]` to throttle traffic.

12.4 Clocking and Reset

Delta uses a single clock (`aclk`) and an active-low asynchronous reset (`aresetn`). All registered state is reset; combinational decode and mux logic is not registered.

12.5 Relationship to Other Components

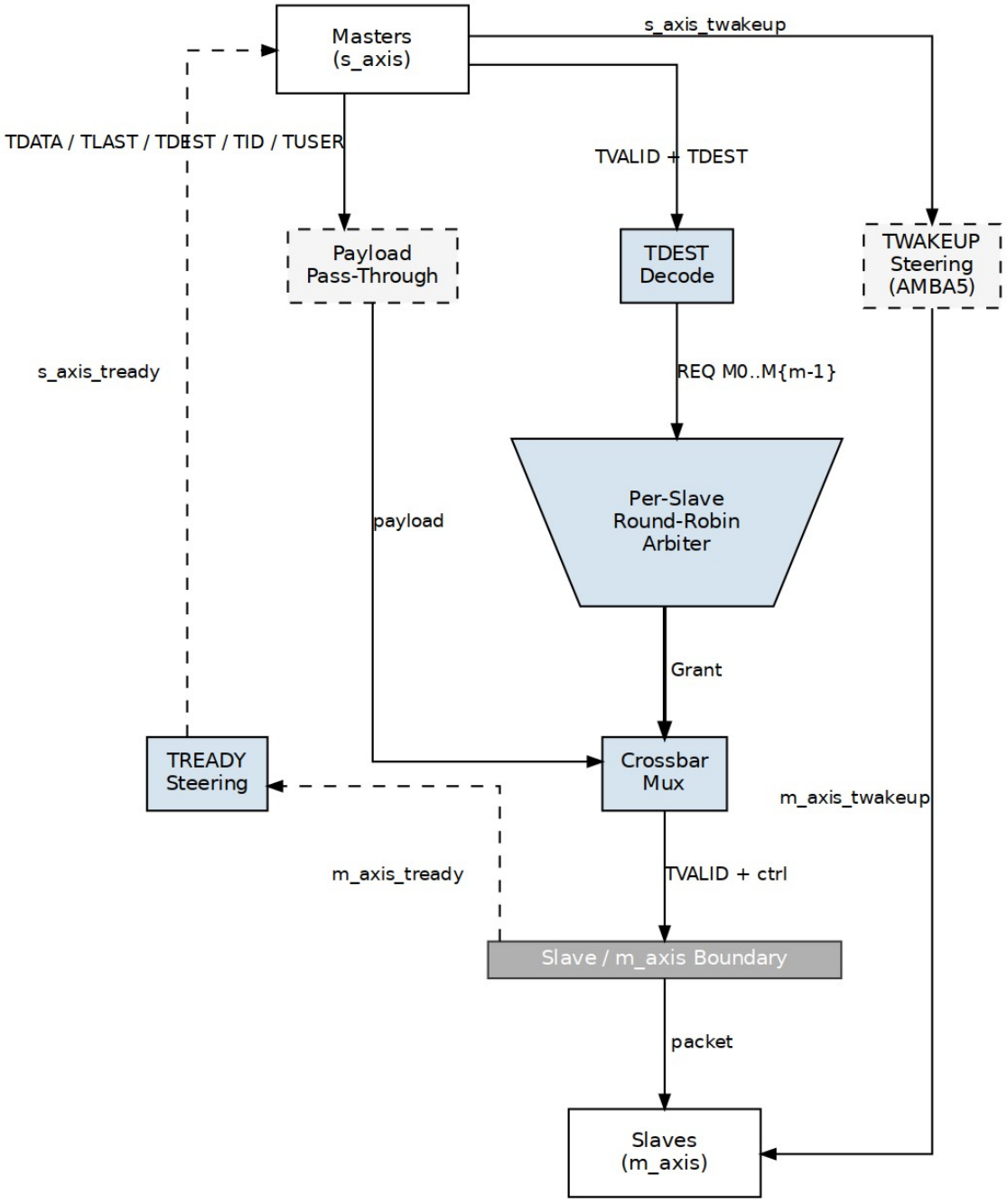
Component	Role	When to Use
Delta	AXI4-Stream crossbar generator	Streaming fabrics, DSP arrays, sensors
Bridge	Multi-protocol crossbar generator	Memory-mapped AXI4/AXI4-Lite/APB systems
APB Crossbar	APB-only crossbar	Low-bandwidth register-mapped peripherals
RAPIDS	DMA subsystem	Descriptor-based memory transfers

Delta is intentionally narrow in scope: it does one thing (streaming crossbars) and does it without protocol conversion.

13 Block Diagram

13.1 Delta Architecture Overview

13.1.1 Figure 3.1: Delta High-Level Block Diagram



Delta HAS Block Diagram

The diagram shows the three traceable flows of the Delta crossbar: the request/control flow down the center, the payload flow on the left, and the backpressure flow returning up the right/left side as a dashed sideband. The scheduler is the wide-edge-up trapezium (per-slave round-robin arbiter). The gray bar is the slave / `m_axis` boundary.

13.2 Functional Blocks

13.2.1 TDEST Decode

Receives `s_axis_tvalid` and `s_axis_tdest` from every master and produces the per-slave `request_matrix`. A request bit is set when a master has a valid transaction whose TDEST selects that slave.

13.2.2 Per-Slave Round-Robin Arbiter

One arbiter per slave. When the slave is idle, it picks the next requesting master in round-robin order. Once a master is granted, the grant is held until TLAST is accepted. This block is the scheduler analog in the house style and is drawn as an inverted trapezium.

13.2.3 Crossbar Mux

A bank of per-slave multiplexers. Each slave mux selects the payload and control signals from the master that currently holds the grant for that slave.

13.2.4 TREADY Steering

Steers each slave's `m_axis_tready` back to the single master that currently owns the grant for that slave. Ungranted masters see `s_axis_tready == 0`.

13.2.5 TWAKEUP Steering (AMBA5, Optional)

When `--amba5` is enabled, this block ORs each master's `s_axis_twakeup` to the slave indicated by `s_axis_tdest`. Because TWAKEUP is a hint, it does not depend on the grant and may assert before TVALID.

13.2.6 Packet Counters (Optional)

32-bit counters per master and per slave that increment on every completed packet (`TVALID && TREADY && TLAST`). Removed from the build with `--no-counters`.

13.3 Scalability

Table 3.1: Topology Parameter Scaling

Parameter	Typical Range	Impact
NUM_MASTERS	1-32	Linear arbitration and mux growth
NUM_SLAVES	1-256	Linear decode and grant-matrix growth
DATA_WIDTH	8-1024	Linear data-path width
DEST_WIDTH	auto (`clog2(NUM_SLAVES)`)	Sets TDEST width
ID_WIDTH	auto (`clog2(NUM_MASTERS)`)	Sets TID width

13.3.1 Resource Scaling

- **Logic:** O(NUM_MASTERS × NUM_SLAVES) for the grant and request matrices.
- **Data path:** O(NUM_SLAVES × DATA_WIDTH) for the output muxes.
- **Memory:** No BRAM; all state is flip-flops.

14 Data Flow

14.1 Request Flow

1. A master asserts `s_axis_tvalid[m]` and places the target slave index on `s_axis_tdest[m]`.
2. The TDEST decode block sets `request_matrix[s][m]` for the selected slave.
3. If the slave is not already serving another packet, the per-slave round-robin arbiter grants one requesting master.
4. The grant is registered and held until the packet ends on `TLAST`.

14.2 Payload Flow

1. While a grant is active, the crossbar mux selects the granted master's payload.
2. `m_axis_tdata[s]`, `m_axis_tlast[s]`, `m_axis_tdest[s]`, `m_axis_tid[s]`, and `m_axis_tuser[s]` are driven from the granted master.
3. If `TSTRB` is enabled, `m_axis_tstrb[s]` is also forwarded.
4. If `TPARITY` is enabled, `m_axis_tparity[s]` is forwarded.

14.3 Backpressure Flow

1. Each downstream slave drives `m_axis_tready[s]`.
2. The TREADY steering block routes the granted slave's ready back to the owning master.
3. Only the granted master sees `s_axis_tready[m] == 1`; all other masters are stalled.

14.4 Wakeup Flow (AMBA5)

1. A master may assert `s_axis_twakeup[m]` before asserting `s_axis_tvalid[m]`.
2. The TWAKEUP steering block decodes `s_axis_tdest[m]` and ORs the hint to the matching slave.
3. `m_axis_twakeup[s]` may assert with no active grant.

14.5 Packet Boundaries

TLAST marks the final beat of a packet. The arbiter releases the grant on the cycle in which `m_axis_tvalid[s] && m_axis_tready[s] && m_axis_tlast[s]` is true. The next arbitration decision can occur on the following cycle.

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15 Protocol Support

15.1 AXI4-Stream Base Flavor

The default generated crossbar implements the standard AXI4-Stream signal set:

- `TDATA` — data payload
- `TVALID` / `TREADY` — handshake
- `TLAST` — packet boundary
- `TDEST` — destination slave index
- `TID` — transaction/master ID (pass-through)
- `TUSER` — user sideband (pass-through)

This is the byte-identical default; no sideband options are enabled unless requested.

15.2 Optional TSTRB Byte Qualifiers

The `--strb-width N` option adds TSTRB to every master and slave interface. TSTRB is treated as a pass-through sideband: the mux forwards the granted master's TSTRB to the selected slave. Delta does not interpret or modify strobe bits.

15.3 Optional AMBA 5 Sidebands

The `--amba5` option adds two AMBA 5 signals:

- `TWAKEUP` — wakeup hint, decoded from `TDEST` and ORed per slave.
- `TPARITY` — parity sideband, treated as pass-through like `TUSER`.

The parity width defaults to `DATA_WIDTH / 8` and can be overridden with `--parity-width M`. The generated ports port-match the `rtl/amba/axis5/*` conventions.

15.4 What Delta Does Not Support

Delta is intentionally AXI4-Stream only. It does **not** support:

- AXI4 memory-mapped transactions (AW/AR/W/B/R channels).
- AXI4-Lite register-mapped access.
- APB transfers.
- Burst lengths, address phases, or memory attributes.
- Dynamic reconfiguration of the crossbar dimensions.

For memory-mapped fabrics, use the sibling `bridge` or `apbx-xbar` components.

15.5 Protocol Compliance Summary

Table 3.2: Protocol Support Matrix

Feature	Base Flavor	Rich Flavor	Notes
TVALID/TREADY handshake	Yes	Yes	Standard AXI4-Stream
TLAST packet boundary	Yes	Yes	Grant release on TLAST
TDEST routing	Yes	Yes	Direct slave index
TID/TUSER pass-through	Yes	Yes	Unmodified
TSTRB	No	Yes	`--strb-width N`
TWAKEUP	No	Yes	`--amba5`
TPARITY	No	Yes	`--amba5 [--parity-width M]`

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16 AXI-Stream Base Interface

16.1 Master Interface (S_AXIS)

Each master port is indexed by `m` in the range `0` to `NUM_MASTERS-1`.

Table 4.1: Base Master Interface Signals

Signal	Width	Direction	Description
`s_axis_tdata[m]`	`DATA_WIDTH`	Input	Data payload.
`s_axis_tvalid[m]`	1	Input	Valid indicator. When high, `s_axis_tdest[m]` selects the target slave.
`s_axis_tready[m]`	1	Output	Ready / backpressure. Driven from the granted slave's `m_axis_tready`.
`s_axis_tlast[m]`	1	Input	Packet boundary. The final beat of a packet.
`s_axis_tdest[m]`	`DEST_WIDTH`	Input	Target slave index. Must be `< NUM_SLAVES` when `tvalid` is asserted.
`s_axis_tid[m]`	`ID_WIDTH`	Input	Transaction ID. Pass-through to `m_axis_tid`.
`s_axis_tuser[m]`	`USER_WIDTH`	Input	User sideband. Pass-through to `m_axis_tuser`.

16.2 Slave Interface (M_AXIS)

Each slave port is indexed by `s` in the range `0` to `NUM_SLAVES-1`.

Table 4.2: Base Slave Interface Signals

Signal	Width	Direction	Description
`m_axis_tdata[s]`	`DATA_WIDTH`	Output	Data payload from the granted master.
`m_axis_tvalid[s]`	1	Output	Valid indicator. Asserted when a grant is active and the selected master is valid.
`m_axis_tready[s]`	1	Input	Ready / backpressure from the downstream slave.
`m_axis_tlast[s]`	1	Output	Packet boundary. Forwarded from the granted master.
`m_axis_tdest[s]`	`DEST_WIDTH`	Output	Target slave index. Pass-through from the granted master.
`m_axis_tid[s]`	`ID_WIDTH`	Output	Transaction ID. Pass-through from the granted master.
`m_axis_tuser[s]`	`USER_WIDTH`	Output	User sideband. Pass-through from the granted master.

16.3 Signal Semantics

- `TVALID` may be asserted without `TREADY`; the transfer completes only when both are high.
- `TREADY` for an ungranted master is always low.
- `TDEST` must be stable while `TVALID` is asserted and the master is waiting for grant or transfer.
- `TID` and `TUSER` are pass-through and must be stable on every valid beat.

17 AXI-Stream Rich Interface (TSTRB + AMBA5)

17.1 Rich Master Interface (S_AXIS)

When `--strb-width N` and/or `--amba5` are used, the master interface adds the following signals in addition to the base set.

Table 4.3: Rich Master Interface Additional Signals

Signal	Width	Direction	Presence	Description
`s_axis_tstrb[m]`	`STRB_WIDTH`	Input	`--strb-width N`	Byte qualifier / strobe. Pass-through to `m_axis_tstrb`.
`s_axis_twakeup[m]`	1	Input	`--amba5`	Wakeup hint. May assert before `TVALID`.
`s_axis_tparity[m]`	`PW_WIDTH`	Input	`--amba5`	Parity sideband. Pass-through to `m_axis_tparity`.

17.2 Rich Slave Interface (M_AXIS)

Table 4.4: Rich Slave Interface Additional Signals

Signal	Width	Direction	Presence	Description
m_axis_tstrb[s]	STRB_WIDTH	Output	--strb-width N	Byte qualifier from the granted master.
m_axis_twakeup[s]	1	Output	--amba5	Wakeup hint. OR of all masters whose TDEST selects slave s.
m_axis_tparity[s]	PW_WIDTH	Output	--amba5	Parity sideband from the granted master.

17.3 TSTRB Semantics

- TSTRB is treated as an opaque byte-qualifier sideband.
- The crossbar forwards the granted master's TSTRB to the selected slave.
- Delta does not gate TVALID based on TSTRB or modify strobe bits.

17.4 TWAKEUP Semantics

TWAKEUP is an AMBA 5 hint, not a handshake signal:

- A master may assert `s_axis_twakeup[m]` before `s_axis_tvalid[m]` to wake the target slave.
- The crossbar decodes `s_axis_tdest[m]` and ORs the hint to the indicated slave.
- `m_axis_twakeup[s]` may be high while `m_axis_tvalid[s]` is low.
- No grant is required for TWAKEUP steering.

17.5 TPARITY Semantics

- `TPARITY` is treated as a pass-through sideband like `TUSER`.
- The width defaults to `DATA_WIDTH / 8` and can be overridden with `--parity-width M`.
- The crossbar forwards the granted master's `TPARITY` to the selected slave.

17.6 Parameter Implications

Table 4.5: Rich-Flavor Parameter-to-Port Mapping

Option	Generated Parameter	Port Effect
`--strb-width N`	`STRB_WIDTH = N`	Adds `s_axis_tstrb` and `m_axis_tstrb` arrays.
`--amba5`	`PW_WIDTH = DATA_WIDTH/8` (default)	Adds `s_axis_twakeup/tparity` and `m_axis_twakeup/tparity` arrays.
`--amba5 --parity-width M`	`PW_WIDTH = M`	Overrides the default parity width.

18 Clock and Reset

18.1 Clock

Signal	Width	Direction	Description
`aclk`	1	Input	Positive-edge clock for all registered state.

All registered logic (grants, `last_grant`, packet-active flags, counters) is clocked on the rising edge of `aclk`. Combinational decode and mux logic is not registered.

18.2 Reset

Signal	Width	Direction	Description
`aresetn`	1	Input	Active-low asynchronous reset.

`aresetn` resets all registered state:

- `grant_matrix` to zero.
- `last_grant` to zero.
- `packet_active` to zero.
- Packet counters to zero.

The reset assertion is asynchronous; de-assertion is synchronous to `aclk` in the checked-in RTL through the `ALWAYS_FF_RST` macro.

18.3 Clocking Rules

- All masters and slaves share the same `aclk`.
- Delta does not contain clock-domain crossing logic.
- Inputs are sampled on the rising edge of `aclk`; outputs are driven combinationally or from registers.

18.4 Reset Recommendations

- Assert `aresetn` for at least several clock cycles after power and clock stability.
- Do not rely on Delta's outputs during reset; combinational mux outputs may be zero or undefined during reset.

---

19 Throughput Characteristics

19.1 Aggregate Throughput

Because each slave has its own arbiter and mux, aggregate throughput scales with the number of slaves. In the best case — every master targets a different slave and every slave is ready — the crossbar can accept one beat per master per cycle.

19.2 Per-Slave Throughput

When multiple masters contend for the same slave, the round-robin arbiter gives each contending master an equal share of the slave's bandwidth. With K contending masters, each sees approximately 1/K of the slave's available bandwidth.

19.3 Head-of-Line Blocking

A master blocked on one slave does not block traffic to other slaves. Each slave arbiter is independent. However, a master that targets a heavily contended slave will wait until its round-robin turn and until the current packet completes.

19.4 Throughput Targets

Table 5.1: Throughput Targets

Metric	Target	Notes
Per-slave throughput	Up to 1 transfer/cycle	When one master continuously feeds one ready slave.
Aggregate throughput	Up to NUM_MASTERS transfers/cycle	When all masters target different ready slaves.
Mixed-traffic throughput	≥ 0.7 transfers/cycle per active master	Under moderate contention in the performance model.

19.5 Factors That Reduce Throughput

- Contention for the same slave.
- Downstream backpressure (`m_axis_tready[s] == 0`).
- Multi-beat packets that lock the slave until `TLAST`.
- Arbitration overhead of one cycle when switching masters after `TLAST`.

---

20 Latency Analysis

20.1 Cycle Breakdown

The generated flat crossbar has a fixed two-cycle latency from master input to slave output:

Table 5.2: Latency Cycle Breakdown

Cycle	Action
0	Master asserts `s_axis_tvalid` and `s_axis_tdest`.
1	TDEST decode produces `request_matrix`; arbiter registers the grant.
2	Crossbar mux drives `m_axis_tdata`, `m_axis_tvalid`, etc. to the slave.

20.2 Arbitration Delay

If the target slave is idle when the request arrives, the grant is registered on the next rising edge of `aclk`. If the slave is busy serving another packet, the new request waits until the current packet's TLAST is accepted.

20.3 Backpressure Latency

`m_axis_tready[s]` propagates combinationally through the TREADY steering logic to `s_axis_tready[m]`. There is no additional registered delay on the backpressure path.

20.4 Latency Targets

Table 5.3: Latency Targets

Configuration	Latency	Notes
Flat crossbar, idle slave	2 cycles	Arbitration + registered grant.
Flat crossbar, contended slave	2 + wait cycles	Wait depends on current packet length.

20.5 Frequency Target

The flat crossbar is intended to run at **≥ 300 MHz** on Xilinx UltraScale+ devices for typical configurations (e.g., 4×16 @ 64-bit). Actual Fmax depends on device, tool settings, and floorplanning.

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21 Resources

21.1 Estimates from the Performance Model

The `bin/delta_performance_model.py` script provides empirical resource estimates based on the flat topology. Values below are planning estimates, not synthesis results.

Table 5.4: Resource Estimates

Configuration	LUTs	FFs	Notes
4×16 @ 64-bit	~1,536	~1,536	Base flavor, with counters.
2×4 @ 32-bit	~200	~150	Small test configuration.
4×16 @ 64-bit, no counters	~1,400	~1,200	`--no-counters` saves FF area.

21.2 Scaling Laws

- **LUTs:** Grow roughly as `NUM_MASTERS × NUM_SLAVES` for arbitration plus `NUM_SLAVES × DATA_WIDTH` for the data mux.
- **FFs:** Grow roughly as `NUM_MASTERS × NUM_SLAVES` for the grant matrix and `last_grant` storage, plus `NUM_MASTERS + NUM_SLAVES` for packet counters when enabled.
- **BRAM:** None. Delta is a pure-logic implementation.

21.3 Area Optimization Knobs

Table 5.5: Area Optimization Options

Knob	Effect
`--no-counters`	Removes `pkt_count_master` and `pkt_count_slave` arrays.
Smaller `NUM_MASTERS`/`NUM_SLAVES`	Linear reduction in control logic.
Smaller `DATA_WIDTH`	Linear reduction in data mux area.
Disable richness options	Removes `TSTRB`/`TPARITY` muxing and `TWAKEUP` decode logic.

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22 System Requirements

22.1 Toolchain Requirements

- **Synthesis:** Any SystemVerilog-2017-capable synthesis tool (Vivado, Yosys, Genus, etc.).
- **Simulation:** Any SystemVerilog simulator; cocotb is the planned verification framework.
- **Lint:** Verilator (`--lint-only`) is used for quick sanity checks.

22.2 Runtime Requirements

- Single clock domain: all `s_axis_*` and `m_axis_*` interfaces use `aclk`.
- Active-low asynchronous reset: `aresetn` must be asserted at power-up.
- All upstream masters and downstream slaves must implement the AXI4-Stream handshake.

22.3 Interface Assumptions

- Masters drive valid `TDEST` values (`0` to `NUM_SLAVES-1`).
- Slaves drive `m_axis_tready` independently.
- No master changes `TDEST` while waiting for a grant.
- `TID` and `TUSER` widths match the generated parameters.

22.4 Integration Checklist

- ☐ Confirm `DATA_WIDTH`, `NUM_MASTERS`, and `NUM_SLAVES` match the system.
- ☐ Confirm `DEST_WIDTH` and `ID_WIDTH` are wide enough for the generated module.
- ☐ Decide whether `TSTRB` or AMBA 5 sidebands are needed.
- ☐ Decide whether packet counters are needed (default on; use `--no-counters` to remove).
- ☐ Connect `aclk` and `aresetn` from the system.
- ☐ Verify downstream slaves can accept backpressure.
- ☐ Run `verilator --lint-only` on the generated RTL.

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23 Parameter Configuration

23.1 Generated Module Parameters

Table 6.1: Generated Module Parameters

Parameter	Default	Description
DATA_WIDTH	CLI --data-width	Width of TDATA in bits.
DEST_WIDTH	clog2(NUM_SLAVES)	Width of TDEST.
ID_WIDTH	clog2(NUM_MASTERS)	Width of TID.
USER_WIDTH	CLI --user-width	Width of TUSER.
NUM_MASTERS	CLI --masters	Number of master interfaces.
NUM_SLAVES	CLI --slaves	Number of slave interfaces.
STRB_WIDTH	CLI --strb-width	Width of TSTRB (only when --strb-width is used).
PW_WIDTH	DATA_WIDTH/8 (or --parity-width)	Width of TPARITY (only when --amba5 is used).

23.2 Generator CLI Options

Table 6.2: Generator Command-Line Options

Option	Values	Default	Description
`--topology`	`flat` only	`flat`	Crossbar topology (tree/both retired 2026-10-12).
`--masters`	1-32	required	Number of master interfaces.
`--slaves`	1-256	required	Number of slave interfaces.
`--data-width`	8-1024	64	`TDATA` width in bits.
`--user-width`	1-32	1	`TUSER` width in bits.
`--strb-width`	≥ 0	0	`TSTRB` width; 0 omits `TSTRB`.
`--amba5`	flag	off	Emit AMBA 5 `TWAKEUP` and `TPARITY`.
`--parity-width`	≥ 0	0	Override `TPARITY` width (requires `--amba5`).
`--no-counters`	flag	off	Disable packet counters.
`--output-dir`	path	`../rtl`	Output directory for generated SystemVerilog.

23.3 Example Configurations

23.3.1 Base flavor 4×16 @ 64-bit

```
python3 bin/delta_generator.py \
    --topology flat --masters 4 --slaves 16 --data-width 64 \
    --output-dir rtl/
```

23.3.2 Rich flavor 2×4 @ 32-bit with TSTRB and AMBA5

```
python3 bin/delta_generator.py \
    --topology flat --masters 2 --slaves 4 --data-width 32 \
    --strb-width 4 --amba5 --output-dir rtl_test/
```

23.3.3 Base flavor without counters

```
python3 bin/delta_generator.py \
    --topology flat --masters 4 --slaves 16 --data-width 64 \
    --no-counters --output-dir rtl/
```

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24 Verification Strategy

24.1 Current Status

As of 2026-10-12, **there is no `dv/` test suite for Delta**. The generator output is verified by linting (Verilator) and by visual/code review. A cocotb-based testbench is the known TODO.

24.2 Generator Verification

- Run `bin/delta_generator.py` for several configurations.
- Compare output against checked-in reference RTL (`rtl/delta_axis_flat_4x16.sv`, `rtl_test/delta_axis_flat_*.sv`).
- Verify that default output is byte-identical to pre-2026-10-12 generations.

24.3 Lint Verification

```
verilator --lint-only rtl/delta_axis_flat_4x16.sv
```

Or use the provided `rtl/Makefile`:

```
cd rtl && make verilator
```

24.4 Planned RTL Verification

When a cocotb testbench is added, the recommended coverage includes:

- All master-to-slave routing combinations.
- Concurrent traffic to the same slave (round-robin fairness).
- Concurrent traffic to different slaves (no crosstalk).
- Backpressure stress tests.
- Packet atomicity verification (no interleaving before `TLAST`).
- Rich-flavor tests (`TSTRB`, `TWAKEUP`, `TPARITY`).
- Reset behavior.

24.5 Formal Properties

When `FORMAL` is defined, the generated RTL includes:

- One-hot grant assertion per slave.
- TDEST range assertion per master.

These can be exercised with a SystemVerilog formal tool.

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25 Synthesis and Implementation

25.1 Synthesis Flow

The generated module is a flat SystemVerilog file intended for standard synthesis:

```
# Example Vivado flow
read_verilog rtl/delta_axis_flat_4x16.sv
synth_design -top delta_axis_flat_4x16 -part xczu7ev-ffvc1156-2-e
```

25.2 Timing Considerations

- The critical path is typically from `s_axis_tvalid/tdest` through the decode and mux to `m_axis_tdata/tvalid`.
- Because the grant is registered, the mux select is stable for a full cycle once a grant is made.
- The `TREADY` path is combinational from `m_axis_tready` to `s_axis_tready`.

25.3 Floorplanning Suggestions

- For large crossbars, consider registering `s_axis_tdata` at the master boundary if the combinational mux path is too long.
- Keep related master/slave groups physically close to reduce routing congestion.
- If Fmax is insufficient, the generator supports `--no-counters` and reduced `DATA_WIDTH` to ease timing.

25.4 Implementation Checklist

- ☐ Synthesize the generated module in the target tool.
- ☐ Check setup/hold timing on `aclk`.
- ☐ Verify `aresetn` deassertion is synchronous to `aclk`.
- ☐ Run a post-implementation functional simulation if the design is modified after generation.